# Predicting churn in the customers of a telecom company

**Business Problem**

Churn, customer turnover, attrition, whatever you might call it, it is a metric of great importance to most companies. Churn is, essentially, the loss of customers or the failure to attract customers (https://www.techtarget.com/searchcustomerexperience/definition/churn-rate). In this project, I will create a predictive model to identify customers who are likely to leave, so that the company can intervene and make an offer that will encourage retaining the customer. Reliable, long-term customers are an important part of maintaining profitability.

**Background/History**

This dataset comes populated from IBM, for use in creating models to predict churn (https://www.kaggle.com/datasets/blastchar/telco-customer-churn/data). This data is fictional, but resembles that of a real telecom company. I have worked with churn datasets before, and understand what information is needed from a data science analysis. The overall goal is to identify predicting features of customers who are likely to leave (become churn). By looking at the profiles of customers who have left in the past month, we can identify differences between them and customers who have stayed with the company for a long time. Identifying points that may cause customers to leave the service, and identifying vulnerable customers early, allows our company to target marketing and offers to those customers. Retaining customers generally means more profit, as you receive the same income without spending on customer acquisition (https://www.netsuite.com/portal/resource/articles/human-resources/customer-churn-analysis.shtml).

**Data Explanation**

This dataset required minimal processing, but some adjustments had to be made to be able to be parsed by a classification model. A dictionary of the data is provided in the Appendix (1.1).

To be parsed by our predictive models, the data must be cleaned and modified. This consisted of a few different steps. First, the customerID column was dropped, as it contains no information relevant to whether a customer leaves or not. Dummy columns were created for PaymentMethod, Contract, and InternetService, which turns the categorical column into a column for each category, populated with a 1 or 0 depending on which category was selected. Other rows were turned into similar boolean columns, from “Yes/No” or any other category that meant no, such as “No internet service”. For our analysis, we simply need to know which services a customer is or is not using. Charges and tenure are converted into numeric datatypes, and every other feature is an integer. There were some missing values in TotalCharges. These were dropped because there are enough datapoints without them.

As further processing, the data concerning tenure, total, and monthly charges are all standardized for the model predictions. This allows the model to not get thrown off by the differences in scale (for example, the magnitude of difference between total charges and monthly charges), and focus on the relative relationships.

The classes were compared as well, showing a class imbalance, which is typical in this kind of dataset. The data comes out to roughly 75% retained customers and 25% leaving customers. This will be kept in mind for scoring the models.

After exploratory data analysis, the data is also split into targets and features, where the target is the "Churn" column, as we want to determine which customers are likely to leave, and the rest of the columns become the features.

**Methods**

For this prediction, we are going to use various classification models to be able to look at each customer and determine who is likely to become part of churn before it happens. Various classification models and parameters are to be used to find the most effective model.

To make the process of hyperparameter tuning more efficient, GridSearchCV is used to cycle through different combinations of parameters for each model. GridSearchCV also splits the data into train and test sets for cross-validation. Despite this, due to potential data leakage, a separate test dataset is split off from the full dataset before cross-validation is performed.

Six different classification models are used for this project, with the intent of casting a wide net to ensure that different approaches are represented. These models are Logistic Regression, Ridge Classifier, K-Neighbors Classifier, Random Forest Classifier, Support Vector Classifier, and Gradient Boosting Classifier. A range of different parameters is used to help determine the best parameters for the data.

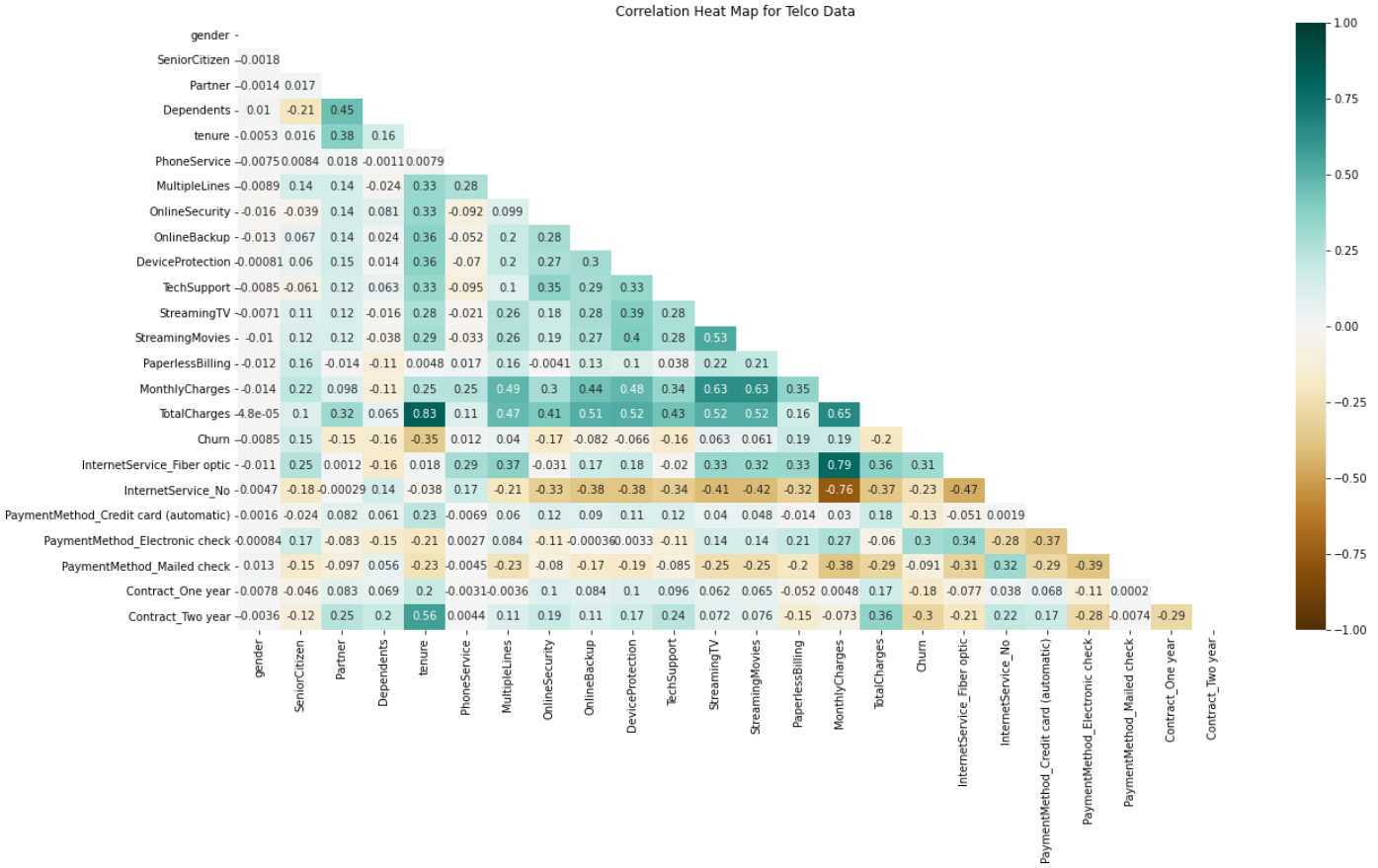
**Analysis**

Two metrics will be used overall to score these models, F1 and accuracy. F1 provides a harmonic mean between precision and recall, which is helpful for unbalanced classes as we are dealing with here. Because there are more customers who stay with the company than leave, our models could easily achieve 75% accuracy by simply predicting that all customers stay. By using F1, accuracy is held in balance with precision, which holds in check false positives and false negatives(https://www.v7labs.com/blog/f1-score-guide).

An F1 score of 1.0 would be ideal, but we need to also determine the amount of false positives or false negatives we can accept. False positives would mean offering discounts to customers who were not considering leaving, while false negatives would mean not identifying a churn candidate and potentially losing a customer.

Because we want to reduce the number of customers leaving as much as possible, accuracy is a metric to keep in mind along with F1. Scoring simply based on accuracy can lead to an abundance of false positives in the predictions. The consequence of false positives in this situation is better than that of a false negative, where deals and offers extended to customers who were not considering leaving, may have the additional effect of increasing brand loyalty. This may count as deposits in a "trust account" with the customer, so when there is a service issue, or competitors are angling for current customers, the loyalty and convenience may keep them from leaving.

A simple correlation heatmap also fails to show a straightforward relationship between churn and other features of the dataset.




**Conclusion**

There is no one overarching trend that defines the customer who is likely to leave the company. Customers who left were more likely to have a month-to-month contract and pay with electronic check. Offering incentives for both options may result in better customer retention over time.

After running GridSearchCV on a number of different classification models and hyperparameters, our Logistic Regression model has given us the highest F1 score, at 0.6, and the accuracy score of 0.8. This is accepted as our top model, as both the F1 and accuracy score are higher than random guessing. These values are double-checked on the remaining test data, for a final score of .598 in accuracy and .799 in F1, in accordance with our previous scores. This model works for our task. Models, best parameters, and scores can be found in the Appendix (1.2).

Using this model, the company can identify customers who are likely to leave soon, and can offer incentives as described below to retain customers.

**Assumptions**

In this dataset, we make assumptions that many of the features are independent of each other (excluding obvious dependencies, like multiple lines and phone service). After exploratory data analysis,

we see that there is generally not much of a difference in the distribution between churn and retained customers when it comes to the different features.

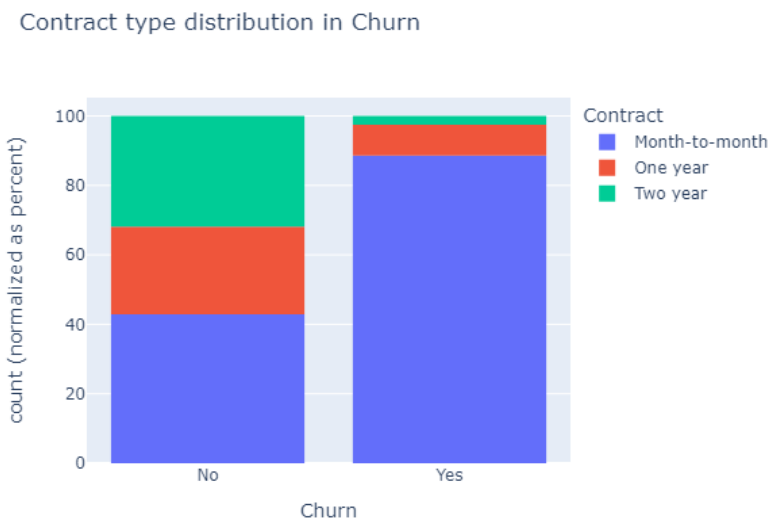
**Limitations**

The main limitations of this dataset are primarily that there are only so many datapoints and features available. A wider range of features, such as more in-depth demographic information, may provide more insight about customer activity and preferences. Additionally, the current dataset shows only one point in time, looking at customers lost within the previous month. This does not tell us what previous lost customers may have looked like, and what trends may be present.

**Challenges**

There were not a lot of challenges with this dataset. The data was already clean and relevant to the project. The biggest challenge is simply that the data is only looking at the recent month of lost customers.

**Future Uses/Additional Applications**

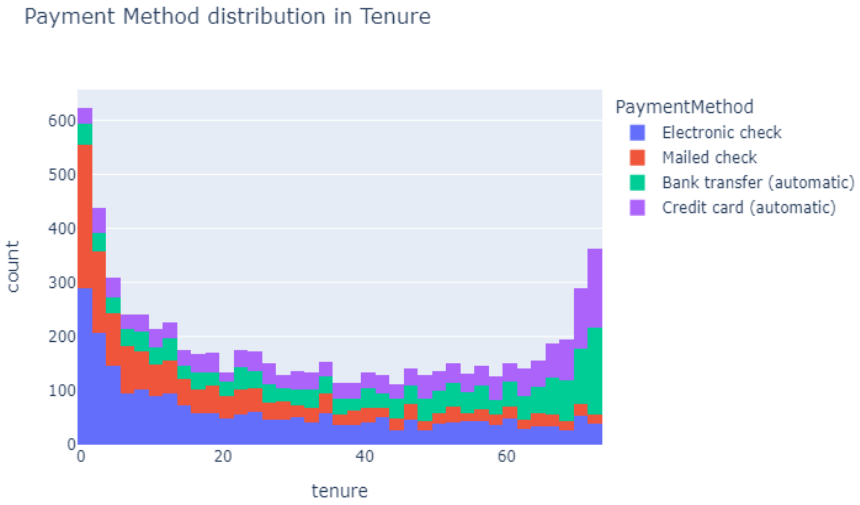



Information gathered from this project can help identify future loss risks, and identify customer wants. This also allows internal promotions and marketing to be deployed strategically to keep customers subscribed. This project has also done most of the work to create future iterations of a predictive model, should new features be introduced, or if the focus were to switch from churn to other offered features by the company, such as Online Backup.

**Recommendations**

Based on the information we are given, certain offers should be made to customers who are likely to churn. Because many customers who leave use electronic check to pay, an incentive to switch payment to a different method may help retain those customers. An overwhelming amount of churn

happens within those customers who use month-to-month plans, so discounts and offers should be made to encourage those customers to switch to a longer contract plan, ideally a 2-year plan.




**Implementation plan**

Using the insights gathered through the exploratory data analysis, and by using the prediction model, specific marketing and promotional campaigns can be deployed to reduce churn and increase subscriber longevity. This model should be implemented on a regular basis (eg: monthly, quarterly, etc) to determine current consumer trends, and to reach out to customers before they begin the process of cancelling their subscriptions.

**Ethical Assessment**

Regarding ethics, this project has little to worry about. There is no personally identifying information involved, and the insights collected are mostly useless outside of the telecom company who collected them. Additionally, this data is fictitious, so there is no tie to real people at all, except by coincidence.

# Appendix

## 1.1 Dictionary of Columns in Telco Dataset

| | |
|---|---|
| customerID | Unique identifier of each customer |
| gender | Male/Female |
| SeniorCitizen | Yes/No - is the subscriber a senior citizen |
| Partner | Yes/No - does the subscriber have a partner |
| Dependents | Yes/No - does the subscriber have dependents |
| tenure | Number of subscribed months |
| PhoneService | Yes/No - does the subscriber have phone service |
| Multiple Lines | Yes/No/No phone service - does the subscriber have multiple lines, or no phone service |
| InternetService | DSL/Fiber Optic/No - which internet service does the subscriber use, if any |
| OnlineSecurity | Yes/No/No internet service – does the subscriber opt in to online security features |
| OnlineBackup | Yes/No/No internet service – does the subscriber opt in to online backup features |
| DeviceProtection | Yes/No/No internet service – does the subscriber opt in to device protection |
| TechSupport | Yes/No/No internet service – does the subscriber opt in to technical support |
| StreamingTV | Yes/No/No internet service – does the subscriber opt in to streaming TV |
| StreamingMovies | Yes/No/No internet service – does the subscriber opt in to streaming movies |
| Contract | Month-to-month/One year/Two year – length of contract term for the subscriber |
| PaperlessBilling | Yes/No - does the subscriber opt in to paperless billing |
| PaymentMethod | Electronic Check/Mailed Check/Bank Transfer/Credit card – method of payment for the account |
| MonthlyCharges | The amount charged to the customer monthly |
| TotalCharges | Total amount charged to the customer over the span of their subscription |
| Churn | Yes/No -whether the account is defined as churned or not. Churn is defined as an account that was closed within the last month |

## 1.2 Model Scoring

| Model | Parameters | F1 Score | Accuracy Score |
|---|---|---|---|
| LogisticRegression | C: 1.0; Penalty: L2; Solver: newton-cg | 0.593 | 0.805 |
| RidgeClassifier | Alpha: 0.1; Solver: auto | 0.573 | 0.802 |
| KNeighborsClassifier | Metric: Manhattan; N_neighbors: 19, Weights: uniform | 0.593 | 0.796 |

| | | | |
|---|---|---|---|
| SVC (C-Support Vector Classification) | C: 1.0; Gamma: scale; Kernel: poly | 0.561 | 0.802 |
| RandomForest | Max_features: log2; N_estimators: 1000 | 0.542 | 0.792 |
| GradientBoostingClassifier | Learning_rate: 0.01; Max_depth: 3; N_estimators: 1000; Subsample: 0.7 | 0.572 | 0.803 |